FIITJEE

ALL INDIA INTERNAL TEST SERIES

CHEMISTRY, MATHEMATICS & PHYSICS

SET – B

APT - 4 | **080165.2** | **Ai²TS-7**

Time Allotted : 3 Hours | **Maximum Marks: 312**

CLASS XII

INSTRUCTIONS

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

- ✓ Attempt ALL the questions. Answers have to be marked on the OMR sheets.
- ✓ This question paper contains **Three Sections.**
- ✓ **Section – I** is *"Chemistry"*, **Section – II** is *"Mathematics"* and **Section – III** is *"Physics"*.
- ✓ Each Section is further divided into two Parts: **Part – A** & **Part – B**.
- ✓ Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
- ✓ Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **PART-A (01 – 08)** contains 8 Multiple Choice Questions which have Only One Correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.
PART-A (09 – 16) contains 4 Paragraphs. Based upon each paragraph, 2 Multiple Choice Questions have to be answered. Each question has Only One Correct answer and carries **+3 marks** for the correct answer and **–1 mark** for a wrong answer.

(ii) **PART-B (01 – 04)** contains 4 Matrix Match Type Question which have statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. There may be One or More Than One Correct choices. Each question carries **+12 marks** for all correct answer however for each correct row **+3 marks** will be awarded and **–1 mark** for each row matched incorrectly.

Name of Candidate :

Batch ID : **Date of Examination :** / / 2 0 1

Enrolment Number :

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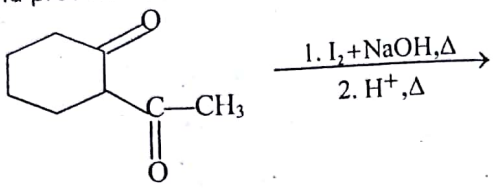
Section – I (CHEMISTRY)

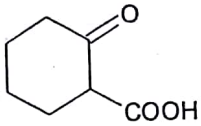
PART – A

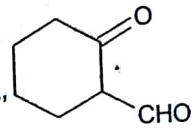
(Single Correct Choice Type)

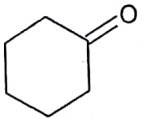
This section contains 8 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

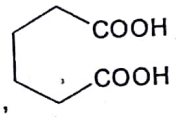
1. End product of the following sequence of reaction is



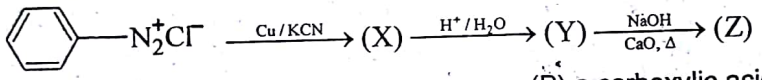
(A) Yellow ppt. of CHI_3, 

(B) Yellow ppt. of CHI_3, 

(C) Yellow ppt. of CHI_3, 

(D) Yellow ppt. of CHI_3, 

2. The end product (Z) of the following reaction is



(A) a cyanide
(B) a carboxylic acid
(C) an amine
(D) an arene

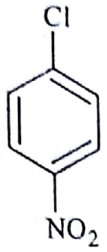
3. An example of a condensation polymer is
(A) PVC
(B) terylene
(C) polypropylene
(D) polystyrene

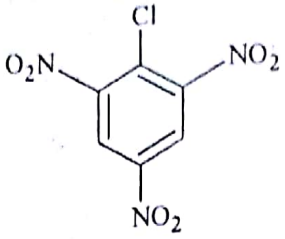
4. Sucrose on hydrolysis yields a mixture which is
(A) Optically inactive
(B) Dextrorotatory
(C) Laevorotatory
(D) Racemic

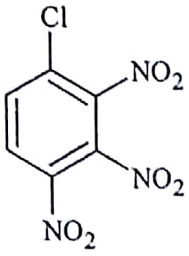
Space for rough work

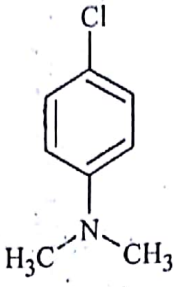
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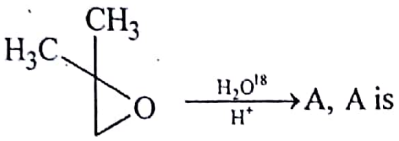
5. Which of the following derivatives of benzene would undergo hydrolysis most readily with aq. KOH

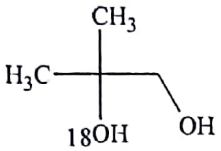
(A) 

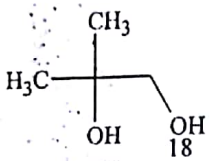
(B) 

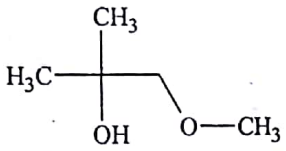
(C) 

(D) 

6. 

(A) 

(B) 

(C) 

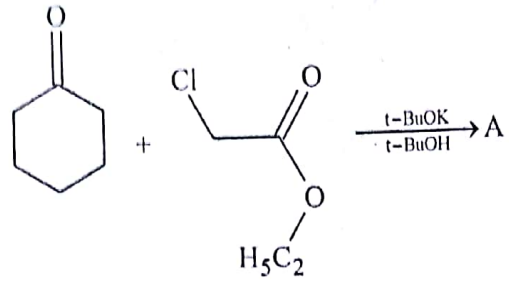
(C) Both are correct

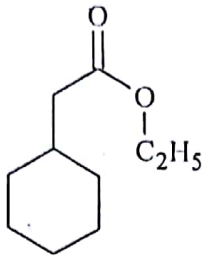
(D) None is correct

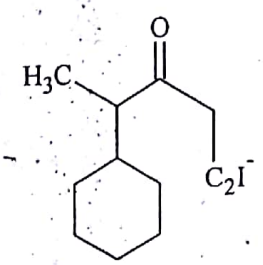
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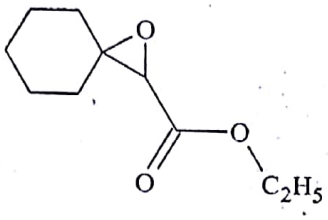
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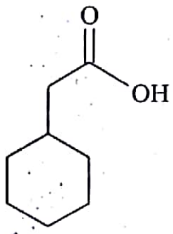
7. What is A in the following reaction?

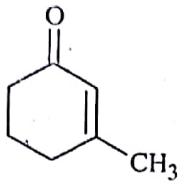


(A) 

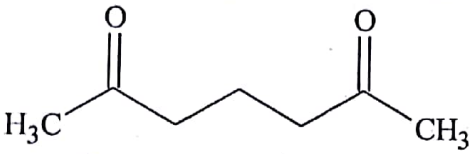
(B) 

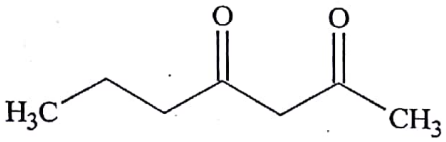
(C) 

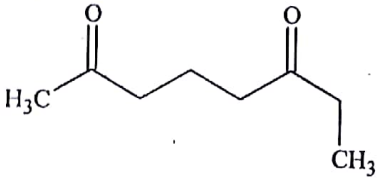
(D) 

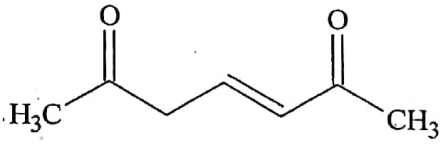
8. 

is the final product obtained when one of the following is reacted with base.

(A) 

(B) 

(C) 

(D) 

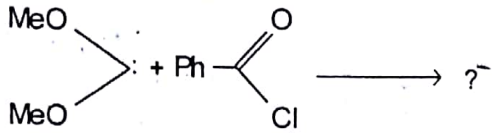
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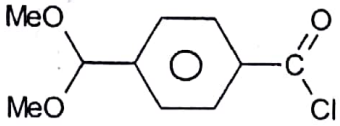
Paragraph
(Single Correct Choice Type)
Paragraph (9 – 10)

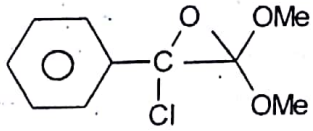
A carbene is a molecule containing a neutral carbon atom with a valence of two and two unshared valence electron. Carbene is important reactive species because carbenes are ambiphilic properties. Some carbenes are aromatic too. Substituents have a profound effect on the reactivity of carbenes, Donor substituents lower the energy more if they are conjugated to empty P-orbital and electron withdrawing substituents lower the energy more if they are conjugated to the filled P-orbital.

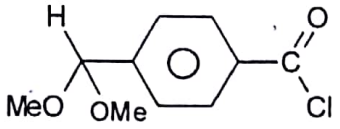
9. In the following reaction

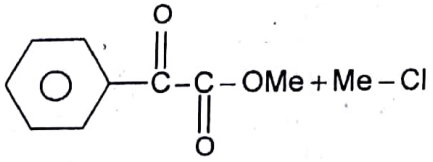


The product(s) is/are

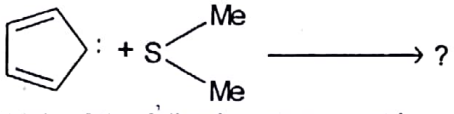
(A) 

(B) 

(C) 

(D) 

10. In the following reaction



Which of the following statement is correct for above reaction condition?

(A) There will be no reaction, since the carbene is aromatic and the reagent is nucleophilic.

(B) An adduct is formed in which carbene acts as nucleophile and reagent acts as electrophile due to availability of vacant d-orbital on central s-atom.

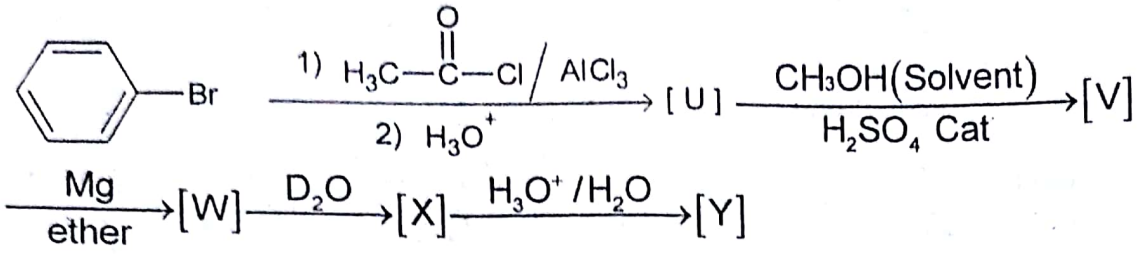
(C) An adduct is formed in which carbene acts as electrophile and thioether acts as nucleophile.

(D) Dimethylthioether undergoes S_N2 reaction

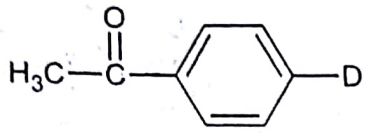
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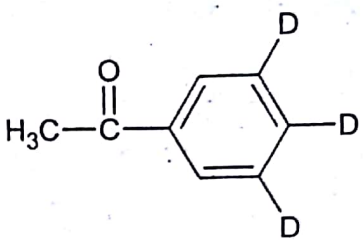
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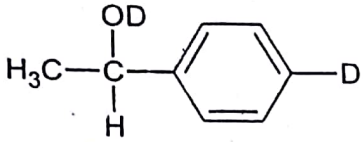
Paragraph (11 – 12)

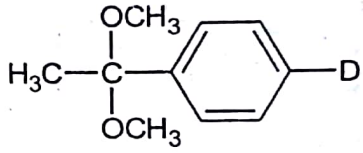


11. The compound [X] is

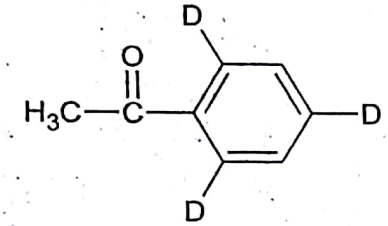
(A) 

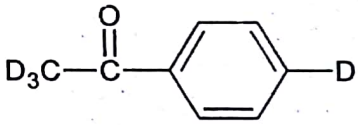
(B) 

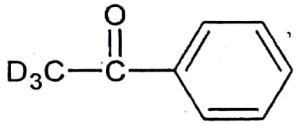
(C) 

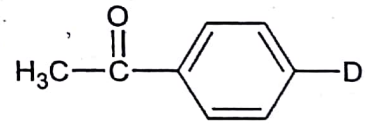
(D) 

12. The compound [Y] is

(A) 

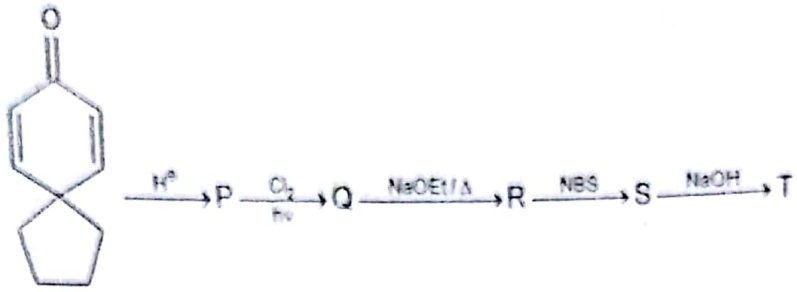
(B) 

(C) 

(D) 

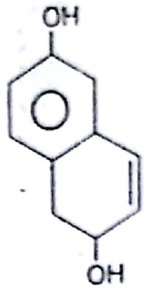
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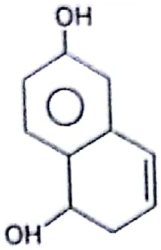
Paragraph (13 – 14)

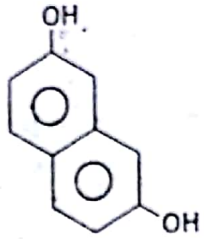


13. Compound 'T' is

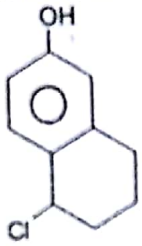
(A) 

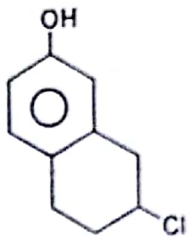
(B) 

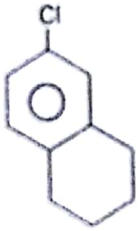
(C) 

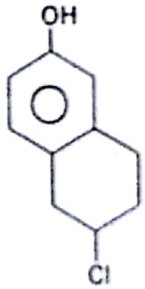
(D) 

14. Compound 'Q' is

(A) 

(B) 

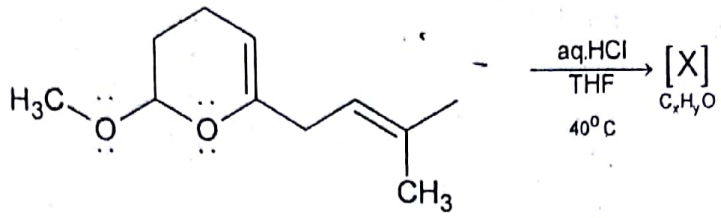
(C) 

(D) 

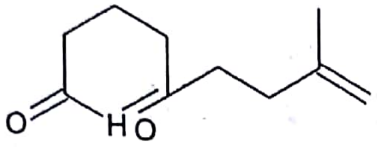
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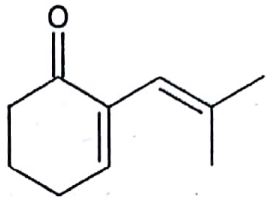
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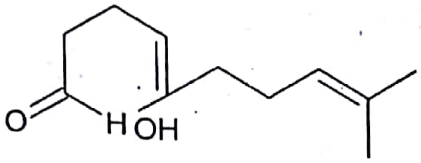
Paragraph (15 – 16)

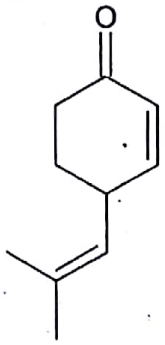


15. The organic compound [X] is

(A) 

(B) 

(C) 

(D) 

16. Which of the following describes the correct sequence of organic reactions during the formation of 'X'?

(A) Acetal hydrolysis first, followed by tautomerisation and then finally acid catalysed aldol condensation.

(B) Acid catalysed hydration first, then hemiacetal and acetal decomposition and finally acid catalyst aldol condensation.

(C) Cleavage of ether first, then acid catalysed hydration and finally acid catalysed aldol condensation.

(D) Acid catalysed hydration first, followed by tautoumerisation and finally acid catalysed alodol condensation.

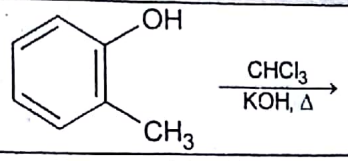
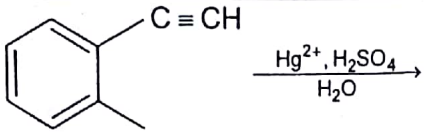
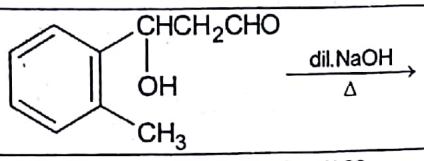
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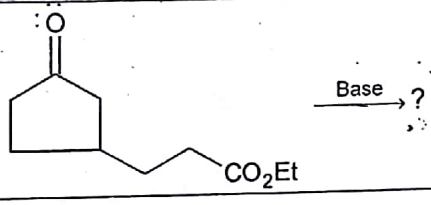
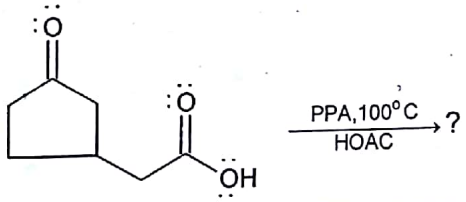
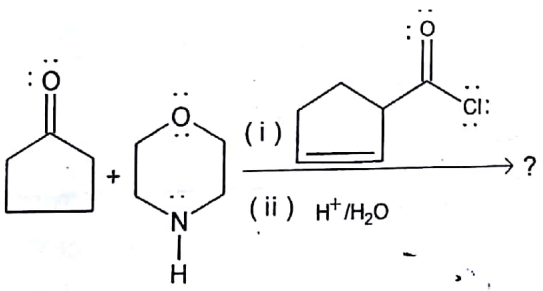
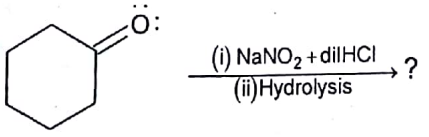
PART – B

(Matrix Match Type)

1. Match the column–I with column–II and mark the correct option from the codes given below:

Column – I		Column – II	
(A)		(P)	Carbocation
(B)		(Q)	Carbanion
(C)		(R)	Carbene
(D)	$(CH_3)_3C-CH_2OH \xrightarrow{Conc.H_2SO_4}$	(S)	Enol

2. Match the columns:

Column – I		Column – II	
(A)		(P)	Cyclic – α – diketone
(B)		(Q)	β – diketone
(C)		(R)	Bicyclic diketone
(D)		(S)	Bridged bicyclic diketone

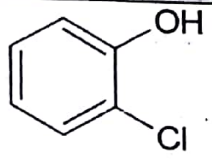
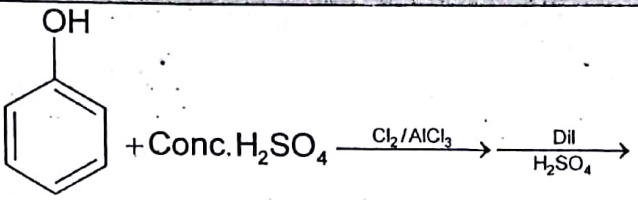
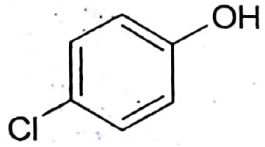
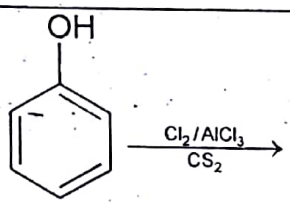
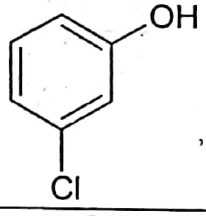
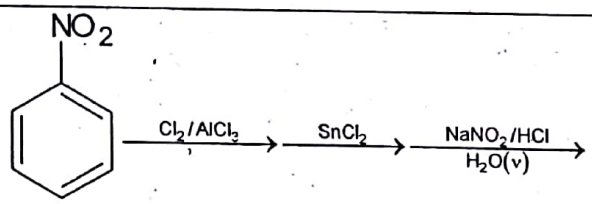
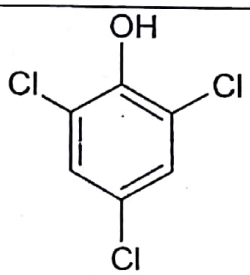
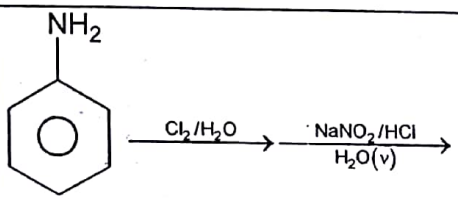
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3. Column-I contains the activated complex of initial reactants and Column-II reflects the effect of increased solvent polarity on reaction rate. Match the column, now

	Column – I		Column – II
(A)	$Y^{\delta+}$ ..... R ..... $X^{\delta+}$	(P)	SN_2, large decrease
(B)	$Y^{\delta-}$ ..... R ..... $X^{\delta+}$	(Q)	E_2, large decrease
(C)	$Y^{\delta+}$ ..... H ..... C – C ..... $X^{\delta-}$	(R)	SN_2, large increase
(D)	$Y^{\delta-}$ ..... H ..... C – C ..... $X^{\delta+}$	(S)	SN_2, small decrease
		(T)	E_2, large increase

4. Match the column–I with column–II and mark the correct option from the codes given below:

	Column – I		Column – II
(A)		(P)	
(B)		(Q)	
(C)		(R)	
(D)		(S)	

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Section – II (MATHEMATICS)

PART – A

(Single Correct Choice Type)

This section contains 8 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. If $b<0$, then the roots x_1 and x_2 of the equation $2x^2+6x+b=0$, satisfy the condition $\left(\frac{x_1}{x_2}\right)+\left(\frac{x_2}{x_1}\right)<k$, then which of the following values can 'k' never exceed.

(A) –3 (B) –5
(C) –6 (D) –2

2. A committee consisting of at least three members is to be formed from a group of 6 boys and 6 girls such that it always has a boy and a girl. The number of ways to form such committee is:

(A) $2^{12}-2^7-13$ (B) $2^{11}-2^6-13$
(C) $2^{12}-2^7-35$ (D) $2^{11}-2^7-35$

3. Let F denote the set of all onto functions from $A=\{a_1,a_2,a_3,a_4\}$ to $B=\{x,y,z\}$. A function f is chosen at random from F. The probability that $f^{-1}\{x\}$ consists of exactly one element is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$
(C) $\frac{1}{6}$ (D) 0

4. Suppose $a,d\in(0,\infty)$ and $a_n=a+(n-1)d\,\forall n\in N$. Least value of $\sum_{k=1}^{n}\sqrt{a_k^2-a_ka_{k+1}+a_{k+1}^2}$ is

(A) $\frac{n}{n+1}\sum_{k=1}^{n+1}a_k$ (B) $\frac{1}{n+1}\sum_{k=1}^{n}a_k$
(C) $\frac{n}{n+1}\sum_{k=1}^{n}a_k$ (D) $\frac{n}{n+1}\sum_{k=2}^{n+1}a_k$

5. Let a, b, c be three distinct complex numbers such that $|a|=|b|=|c|=|b+c-a|=1$, then $3-|b+c|$ is equal to

(A) 1 (B) 2
(C) 3 (D) 4

6. Let $b_i>1$ for $i=1,2,.....,101$. Suppose $\log_e b_1,\log_e b_2,.....,\log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1,a_2,.....a_{101}$ are in A.P. such that $a_1=b_1$ and $a_{51}=b_{51}$. If $t=b_1+b_2+.....+b_{51}$ and $s=a_1+a_2+.....+a_{51}$, then

(A) $s>t$ and $a_{101}>b_{101}$ (B) $s>t$ and $a_{101}<b_{101}$
(C) $s<t$ and $a_{101}>b_{101}$ (D) $s<t$ and $a_{101}<b_{101}$

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7. Suppose m and n are positive integers and let $S=\sum_{k=0}^{n}(-1)^k\frac{1}{k+m+1}\binom{n}{k}$ and

$T=\sum_{k=0}^{m}(-1)^k\frac{1}{k+n+1}\binom{m}{k}$,

then $S-T$ is equal to:

(A) 0
(B) n^m-m^n
(C) $(n+1)^m-(m+1)^n$
(D) $(1-n)^m-(1-m)^n$

8. Let $n\in N$, $S_n=\sum_{r=0}^{3n}{}^{3n}C_r$ and $T_n=\sum_{r=0}^{n}{}^{3n}C_{3r}$, then $|S_n-3T_n|$ equals

(A) 1
(B) 2
(C) 3
(D) 0

Paragraph
(Single Correct Choice Type)
Paragraph (9 – 10)

A polynomial $p(x)=a_0x^n+a_1x^{n-1}+.....+a_{n-1}x+a_n, a_0\neq 0$ is said to be a reciprocal equation if $a_i=a_{n-i}$ for $0\le i\le\left[\frac{n}{2}\right]$ where [x] denote the greatest integer $\le x$.

9. If p(x) is a reciprocal polynomial of odd degree, then one of the roots of $p(x)=0$ is

(A) –1
(B) 1
(C) 0
(D) $\frac{(n+1)}{2}$

10. If p(x) is a reciprocal equation and $a\neq 0$ is a root of $p(x)=0$, then which one of the following is also a root of p(x) = 0?

(A) –a
(B) $\frac{1}{a}$
(C) a – 1
(D) none of these

Paragraph (11 – 12)

Let α,β,γ be three real numbers such that $\alpha^2+\beta^2+\gamma^2-\gamma=0$ and $z=\frac{\alpha+i\beta}{1-\gamma}$.

11. $|z|^2$ equals

(A) γ
(B) $1-\gamma$
(C) $\frac{\gamma}{1-\gamma}$
(D) $\frac{1-\gamma}{\gamma}$

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12. α equals

(A) $\dfrac{z+\bar{z}}{2\left(1+|z|^2\right)}$

(B) $\dfrac{(\bar{z}-z)i}{2\left(1+|z|^2\right)}$

(C) $\dfrac{(z-\bar{z})i}{2\left(1+|z|^2\right)}$

(D) none of these

Paragraph (13 – 14)

Let $a_1, a_2, a_3, \ldots\ldots$ be a harmonic progression such that $a_1 = 5$ and $a_{20} = 25$

13. Value of n for which a_n is maximum is

(A) 22
(B) 23
(C) 24
(D) 25

14. Least value of n for which $a_n < 0$ is

(A) 22
(B) 23
(C) 24
(D) 25

Paragraph (15 – 16)

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}, \frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2, respectively, after two games.

15. $P(X > Y)$ is

(A) $\frac{1}{4}$
(B) $\frac{5}{12}$
(C) $\frac{1}{2}$
(D) $\frac{7}{12}$

16. $P(X = Y)$ is

(A) $\frac{11}{36}$
(B) $\frac{1}{3}$
(C) $\frac{13}{36}$
(D) $\frac{1}{2}$

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PART – B

(Matrix Match Type)

1. Match the equations in column I with the number of solutions in column II

	Column-I		Column-II
(A)	$z^2 + \|z\| = 0$	(P)	1
(B)	$z^2 + \bar{z}^2 = 0$	(Q)	3
(C)	$z^2 + 8\bar{z} = 0$	(R)	4
(D)	$\|z-2\| = 1$ and $\|z-1\| = 2$	(S)	Infinite

2. The greatest integer contained in

	Column-I		Column-II
(A)	$\left(\sqrt{2}+1\right)^{2n}$	(P)	0
(B)	$\left(2-\sqrt{3}\right)^{2n}$	(Q)	is of the form $4k+1$
(C)	$\left(2+\sqrt{3}\right)^{2n}$	(R)	is of the form $2k-1$
(D)	$\left(\sqrt{2}-1\right)^{2n}$	(S)	$2^{n-1}+1$

3. Match the equations in column I with their properties in column II

	Column-I		Column-II
(A)	$a,b,c \in R$ are such that $a+b+c=0$. Equation is $3ax^2+2bx+c=0$	(P)	has real roots
(B)	$a,b,c \in R$ are such that $2a+3b+6c=0$. Equation is $ax^2+bx+c=0$	(Q)	has discriminant > 0
(C)	$a,b,c \in R$ are such that $(a+c)^2 < b^2$. Equation is $ax^2+bx+c=0$	(R)	has at least one root in (0, 1)
(D)	$a,b,c \in R$ are such that $1+\frac{\|b\|}{a}+\frac{c}{a}<0$. Equation $ax^2+bx+c=0$	(S)	has a root in $(-\infty,-1)$ or a root in $(1,\infty)$

4. Letters of the word INDIANOIL are arranged at random. Probability that the word formed

	Column-I		Column-II
(A)	contains the word INDIAN	(P)	$\frac{1}{{}^9C_5}$
(B)	contain the word OIL	(Q)	$\frac{1}{({}^5C_2)({}^7C_2)(9!)}$
(C)	begins with I and ends with L	(R)	$\frac{1}{{}^9C_4}$
(D)	has vowels at the odd places	(S)	$\frac{1}{({}^7C_3)({}^9C_2)}$

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Section – III (PHYSICS)

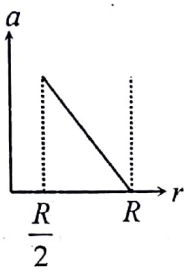
PART – A

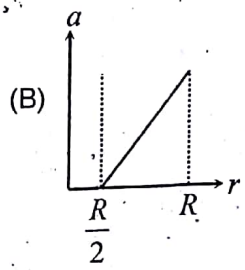
(Single Correct Choice Type)

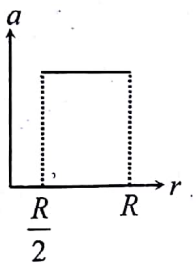
This section contains 8 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

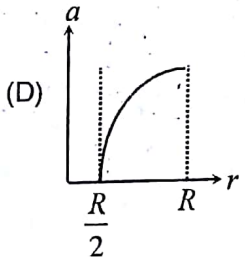
1. Two spherical bodies of masses M and 5M and radii R and 2R respectively are released in free space with initial separation between their centres equal to 12R. If they attract each other due to gravitational force only, then the distance covered by smaller body just before collision is
   (A) 1.5 R (B) 2.5 R
   (C) 4.5 R (D) 7.5 R

2. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is

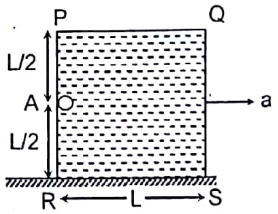
(A) 

(B) 

(C) 

(D) 

3. A small solid ball of density ρ is held inside at point *A* a cubical container of side *L*, filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration *a* horizontally and the ball is released from point *A* simultaneously, then choose the correct alternative.



(A) For ball to hit the top of container at end Q, $a = 3g$

(B) For ball to hit the top of container at end Q, $a = 2g$

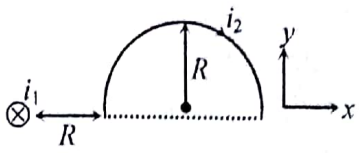
(C) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{L}{g}}$

(D) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{2L}{3g}}$

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4. A very long current carrying wire is placed along z-axis having current of magnitude i_1 towards negative z-axis. A semicircular wire of radius R and having current i_2 is placed in x-y plane, such that line joining two end points of the semicircular wire passes through long wire as shown in figure. Nearest distance of semicircular wire from long wire is R. Net magnetic force on semicircular wire will be

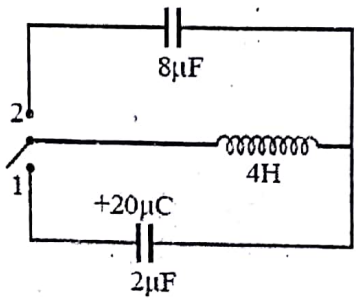


(A) $\frac{\mu_0 i_1 i_2}{2\pi} \ln 3$ (B) $\frac{\mu_0 i_1 i_2}{2\pi} \ln \frac{3}{2}$ (C) zero (D) $\frac{\mu_0 i_1 i_2}{2\pi}$

5. A charge particle enters into a region containing uniform electric field (E) and uniform magnetic field (B) along x-axis and y-axis respectively. If it passes the region unaccelerated, the velocity of charge particle is given by

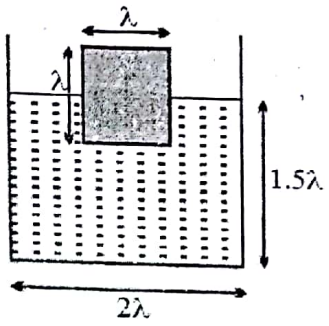
(A) $2\hat{i} + \frac{E}{B}\hat{k}$ (B) $2\hat{j} + \frac{E}{B}\hat{k}$ (C) $-\frac{E}{B}\hat{k}$ (D) none of these

6. At t = 0 two way switch is connected to point 1 as shown in the figure so that current growth start in the LC circuit. At the time $t = t_0$ the switch is shifted towards point 2. Which of the following cannot be the possible value of maximum charge which can be attained by the 8μF capacitor.



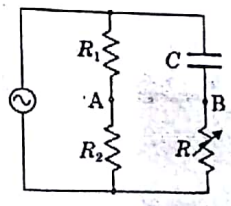
(A) 15 μC
(B) 20 μC
(C) 35 μC
(D) 50 μC

7. A block of side λ is floating half submerged in water of density ρ_w present in a cubical container of side 2λ. Height of free surface of liquid n the container is 1.5λ as shown in the figure. Find of work done by external agent to push the block slowly just inside the liquid.



(A) $\frac{3\lambda^4 \rho_w g}{5}$ (B) $\frac{3\lambda^4 \rho_w g}{32}$
(C) $\frac{3\lambda^4 \rho_w g}{16}$ (D) $\frac{\lambda^4 \rho_w g}{2}$

8. A circuit shown consists of two resistances R_1 and R_2 of equal and fixed values, a variable resistance R, a capacitor of capacitance C and an alternating voltage source. If the variable resistance is made to vary over a wide range, how does the peak voltage between the terminals A and B change?



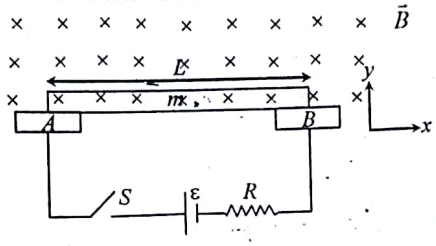
(A) First decreases then increases.
(B) First increases then decreases.
(C) Decreases to a constant value.
(D) Remains constant

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Paragraph
(Single Correct Choice Type)
Paragraph (9 – 10)

In a vertical plane, a metal rod of length L and mass m is placed over two conducting platforms A and B. A region of magnetic field $\vec{B} = -B_0\hat{k}$ starting from line joining A and B and lying over it, is switched on. Now, at $t = 0$ switch S is closed such that the charge q passes through the rod in time dt due to which magnetic field exert an impulsive force which causes the rod to jump with certain velocity and since the gravitational field is present the rod comes back to AB after some time and collides inelastically and the process is repeated



9. The maximum height reached by the rod is

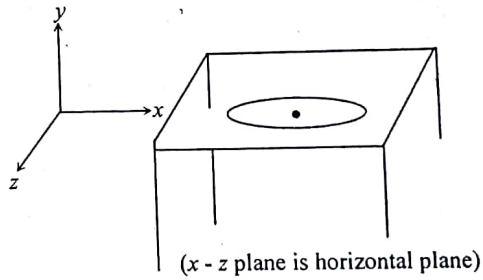
(A) $\frac{2q^2L^2B_0^2}{m^2g}$ (B) $\frac{q^2L^2B_0^2}{2m^2g}$ (C) $\frac{q^2L^2B_0^2}{m^2g}$ (D) $\frac{q^2L^2B_0^2}{4m^2g}$

10. The maximum emf induced in the rod is

(A) $\frac{B_0^2L^2q}{m}$ (B) $\frac{B_0^2L^2q}{2m}$ (C) $\frac{2B_0^2L^2q}{m}$ (D) $\frac{4B_0^2L^2q}{m}$

Paragraph (11 – 12)

A uniform conducting ring of mass π kg and radius 1 m is kept on smooth horizontal table. A uniform but time varying magnetic field $B = (\hat{i} + t^2\hat{j})$ Tesla is present in the region. (Where t is time in seconds). Resistance of ring is 2Ω. Then, ($g = 10\ m/s^2$)



(x - z plane is horizontal plane)

11. Time (in second) at which ring start toppling is

(A) $\frac{10}{\pi}$ (B) $\frac{20}{\pi}$ (C) $\frac{5}{\pi}$ (D) $\frac{25}{\pi}$

12. Heat generated (in kJ) through the ring till the instant when ring start toppling is

(A) $\frac{1}{3\pi}$ (B) $\frac{2}{\pi}$ (C) $\frac{2}{3\pi}$ (D) $\frac{1}{\pi}$

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Paragraph (13 – 14)

A constant current flows through a cylindrical wire of uniform cross-section due to which unit volume of wire generate thermal power P which is transferred to the surroundings through heat conduction. If steady state temperature of surface of wire is T_o

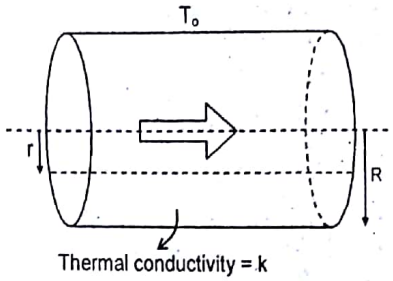
13. Find temperature distribution as a function of r.

(A) $T = T_o + \frac{P}{4k}(R^2 - r^2)$

(B) $T = T_o + \frac{P}{2k}(R^2 - r^2)$

(C) $T = T_o + \frac{P}{2k}(R^2 - r^2)$

(D) $T = T_o + \frac{P}{k}(R^2 - r^2)$



14. If a solid conducting sphere is taken in place of the cylinder (assuming all conditions same as previous question), temperature as function of r will be

(A) $T = T_o + \frac{P}{4k}(R^2 - r^2)$

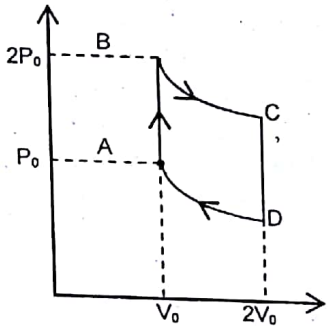
(B) $T = T_o + \frac{P}{6k}(R^2 - r^2)$

(C) $T = T_o + \frac{P}{3k}(R^2 - r^2)$

(D) $T = T_o + \frac{P}{2k}(R^2 - r^2)$

Paragraph (15 – 16)

One mole of a monatomic gas undergoes the cycle shown in the diagram: The pressure, volume and temperature at A are P_0, V_0 and T_0 respectively; the pressure at B is $2P_0$ and the volume at D is $2V_0$. The processes AB and CD are isochoric. During BC DA, the gas undergoes the processes such that during an infinitesimal contraction or expansion, the heat absorbed dQ, the change in internal energy dU and dW, the work done satisfy the relation:



$$dQ = \frac{1}{2}dU + \frac{2}{3}dW$$

For the process BC, DA, we can write

$$dQ = \frac{1}{2}dU + \frac{2}{3}dW = dU + dW$$

or, $+\frac{1}{2}dU = -\frac{1}{3}dW$; or, $\frac{1}{2}\cdot\frac{3}{2}RdT = -\frac{1}{3}dW$

$pdV + Vdp \Rightarrow RdT$; or, $\frac{13}{9}pdV + Vdp = 0$

or, $pV^{\frac{13}{9}} = \text{constant}$

15. Find the temperature at C

(A) $2T_0$

(B) $3T_0$

(C) $2^{5/9}T_0$

(D) $3^{5/9}T_0$

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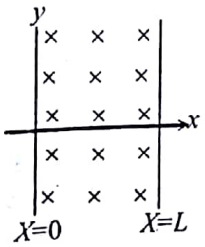
What is the efficiency of the cycle? Assume $2^{5/9} = 1.5, 2^{4/9} = 4/3$ approximately

(A) 24%
(B) 30%
(C) 36%
(D) 42%

PART – B

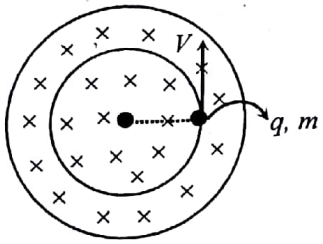
(Matrix Match Type)

1. Magnetic field B_0 is present in the region $X = 0$ to $X = L$ along –ve z-axis. A charge particle having charge q and mass m enters in magnetic along x-axis with speed v_0. If $L = \frac{mv_0}{2qB_0}$, angle of deviation is ϕ_1 and time spent in magnetic field is t_1 and if $L = \frac{\sqrt{3}}{2}\frac{mv_0}{qB_0}$ angle of deviation is ϕ_2 and time spent in magnetic field is t_2, then



Column-I		Column-II	
(A)	ϕ_1 is	(P)	$\frac{\pi m}{6qB}$
(B)	ϕ_2 is	(Q)	$\frac{\pi m}{3qB}$
(C)	t_1 is	(R)	$\pi/3$
(D)	t_2 is	(S)	$\pi/6$
		(T)	$\pi/2$

2. The central cross-section of a long cylindrical region containing uniform but time varying magnetic field B is shown. A particle of constant mass and variable positive charge moves in a circle in the plane, so that the radius of the circle remains constant.



Column-I		Column-II	
(A)	If the magnetic field is increased by 2%, the speed of the particle will	(P)	decrease
(B)	If the magnetic field is decreased by 4%, the speed of the particle will	(Q)	increase
(C)	If the magnetic field is increased by 2%, the charge of the particle will	(R)	change by 1%
(D)	If the magnetic field is decreased by 4%, the charge of the particle will	(S)	change by 2%

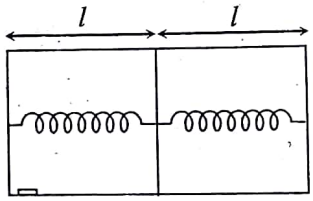
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3. Match the statements in Column I with the result in Column II.

	Column – I		Column – II
(A)	A thin uniform spherical shell of surface area S has an initial temperature more than its surrounding atmosphere. Then magnitude of rate of change of its temperature with time	(P)	Is independent of S
(B)	A soap bubble initially in equilibrium is given a charge Q, which distributes uniformly over its surface. The centre of the bubble is always fixed. For the duration the bubble having surface area S expands the magnitude of electric potential at a fixed point always lying outside the bubble	(Q)	Depends on S
(C)	A container with open top and filed with ideal liquid is placed at rest on a smooth horizontal table. A small hole of area S is drilled at the bottom of a side wall of container. The magnitude of force exerted by escaping liquid on the container	(R)	Remains constant
(D)	An infinitely long straight current carrying wire lies along the axis of a closed cylindrical surface of total surface area S is space. As the magnitude of current in the wire is continuously increased, the magnitude of the magnetic flux through the surface of this cylinder	(S)	Decrease with time

4. A horizontal thermally insulated cylinder of length 2l is separated by a thin insulating piston dividing the cylinder in two equal parts. The piston is connected by ideal springs and initially springs are non-deformed. Left part contains 2 moles of H_2 and right part contains 2 moles of O_2 at same initial temperature. The left part of the cylinder is fitted with a thermostat which maintains the constant temperature of the gas and through which only heat transfer can take place. The following two processes are performed on the given system. After process I, the system is brought back to its initial state.



Process I: Heat Q is supplied to the right part and piston is displaced to the left by a distance l/2.
Process II: Heat Q is given to the right part and as a result piston displaces to the left by a distance l/2. The gas on the left dissociates into its atoms.

	Column-I		Column-II
(A)	In process I	(P)	Temperature of the gas on left remains constant.
(B)	In process I, if piston displaces to the right by same amount.	(Q)	Heat is rejected by the gas on the left.
(C)	In process II	(R)	Total work done by both the gases is positive.
(D)	In process II if partial dissociation has taken place	(S)	Heat is absorbed by the gas on the left.

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